Dynamics Test Answers

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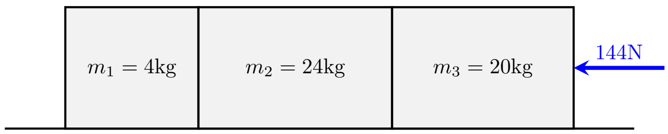
Blocks

1. (6 points) The diagram shows 3 blocks in contact on a smooth surface. A 144N force acts on block 3 from the right, and the surface can be considered frictionless. Calculate:



(a) The acceleration of the system

Solution: The acceleration of the system is $-3\,\mathrm{ms}^{-2}$

(b) The acceleration of each block

Solution: Each block has an acceleration of $-3\,\mathrm{ms}^{-2}$

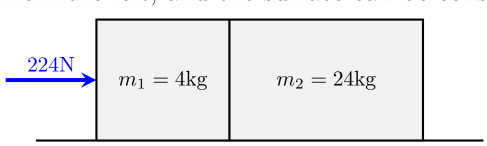
(c) The net force on each block

Solution: Block 1 has a net force of -12N
Block 2 has a net force of -72N
Block 3 has a net force of -60N

(d) The contact forces for each block

Solution: Block 1 has 1 force acting on it. They are:
-12N Contact Force from block 2
Block 2 has 2 forces acting on it. They are:
12N Contact Force from block 1, -84N Contact Force from block 3
Block 3 has 2 forces acting on it. They are:
-144N Applied Force, 84N Contact Force from block 2

2. (6 points) The diagram shows 2 blocks in contact on a smooth surface. A 224N force acts on block 1 from the left, and the surface can be considered frictionless. Calculate:



(a) The acceleration of the system

Solution: The acceleration of the system is $8\,\mathrm{ms}^{-2}$

(b) The acceleration of each block

Solution: Each block has an acceleration of $8\,\mathrm{ms}^{-2}$

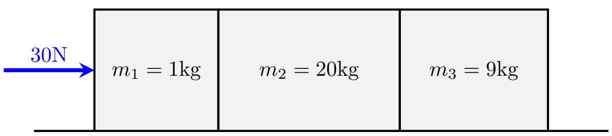
(c) The net force on each block

Solution: Block 1 has a net force of 32N
Block 2 has a net force of 192N

(d) The contact forces for each block

Solution: Block 1 has 2 forces acting on it. They are:
224N Applied Force, -192N Contact Force from block 2
Block 2 has 1 force acting on it. They are:
192N Contact Force from block 1

3. (6 points) The diagram shows 3 blocks in contact on a smooth surface. A 30N force acts on block 1 from the left, and the surface can be considered frictionless. Calculate:



(a) The acceleration of the system

Solution: The acceleration of the system is $1\,\mathrm{ms}^{-2}$

(b) The acceleration of each block

Solution: Each block has an acceleration of $1\,\mathrm{ms}^{-2}$

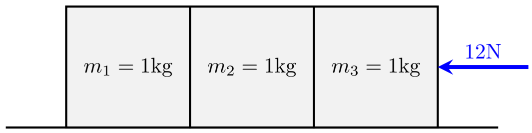
(c) The net force on each block

Solution: Block 1 has a net force of 1N
Block 2 has a net force of 20N
Block 3 has a net force of 9N

(d) The contact forces for each block

Solution: Block 1 has 2 forces acting on it. They are:
30N Applied Force, -29N Contact Force from block 2
Block 2 has 2 forces acting on it. They are:
29N Contact Force from block 1, -9N Contact Force from block 3
Block 3 has 1 force acting on it. They are:
9N Contact Force from block 2

4. (6 points) The diagram shows 3 blocks in contact on a smooth surface. A 12N force acts on block 3 from the right, and the surface can be considered frictionless. Calculate:



(a) The acceleration of the system

Solution: The acceleration of the system is $-4\,\text{ms}^{-2}$

(b) The acceleration of each block

Solution: Each block has an acceleration of $-4\,\text{ms}^{-2}$

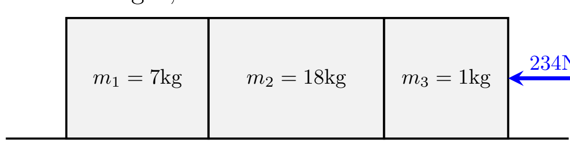
(c) The net force on each block

Solution: Block 1 has a net force of -4N
Block 2 has a net force of -4N
Block 3 has a net force of -4N

(d) The contact forces for each block

Solution: Block 1 has 1 force acting on it. They are:
-4N Contact Force from block 2
Block 2 has 2 forces acting on it. They are:
4N Contact Force from block 1, -8N Contact Force from block 3
Block 3 has 2 forces acting on it. They are:
-12N Applied Force, 8N Contact Force from block 2

5. (6 points) The diagram shows 3 blocks in contact on a smooth surface. A 234N force acts on block 3 from the right, and the surface can be considered frictionless. Calculate:



(a) The acceleration of the system

Solution: The acceleration of the system is $-9\,\text{ms}^{-2}$

(b) The acceleration of each block

Solution: Each block has an acceleration of $-9\,\text{ms}^{-2}$

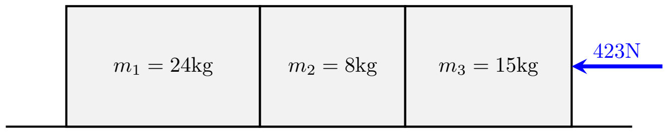
(c) The net force on each block

Solution: Block 1 has a net force of -63N
Block 2 has a net force of -162N
Block 3 has a net force of -9N

(d) The contact forces for each block

Solution: Block 1 has 1 force acting on it. They are:
-63N Contact Force from block 2
Block 2 has 2 forces acting on it. They are:
63N Contact Force from block 1, -225N Contact Force from block 3
Block 3 has 2 forces acting on it. They are:
-234N Applied Force, 225N Contact Force from block 2

6. (6 points) The diagram shows 3 blocks in contact on a smooth surface. A 423N force acts on block 3 from the right, and the surface can be considered frictionless. Calculate:



(a) The acceleration of the system

Solution: The acceleration of the system is $-9\,\mathrm{ms}^{-2}$

(b) The acceleration of each block

Solution: Each block has an acceleration of $-9\,\mathrm{ms}^{-2}$

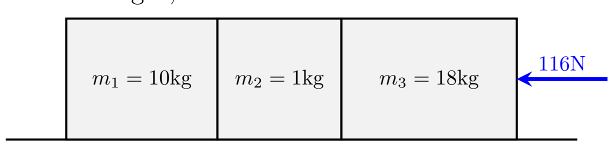
(c) The net force on each block

Solution: Block 1 has a net force of -216N
Block 2 has a net force of -72N
Block 3 has a net force of -135N

(d) The contact forces for each block

Solution: Block 1 has 1 force acting on it. They are:
-216N Contact Force from block 2
Block 2 has 2 forces acting on it. They are:
216N Contact Force from block 1, -288N Contact Force from block 3
Block 3 has 2 forces acting on it. They are:
-423N Applied Force, 288N Contact Force from block 2

7. (6 points) The diagram shows 3 blocks in contact on a smooth surface. A 116N force acts on block 3 from the right, and the surface can be considered frictionless. Calculate:



(a) The acceleration of the system

Solution: The acceleration of the system is $-4\,\mathrm{ms}^{-2}$

(b) The acceleration of each block

Solution: Each block has an acceleration of $-4\,\mathrm{ms}^{-2}$

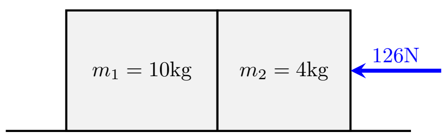
(c) The net force on each block

Solution: Block 1 has a net force of -40N
Block 2 has a net force of -4N
Block 3 has a net force of -72N

(d) The contact forces for each block

Solution: Block 1 has 1 force acting on it. They are:
-40N Contact Force from block 2
Block 2 has 2 forces acting on it. They are:
40N Contact Force from block 1, -44N Contact Force from block 3
Block 3 has 2 forces acting on it. They are:
-116N Applied Force, 44N Contact Force from block 2

8. (6 points) The diagram shows 2 blocks in contact on a smooth surface. A 126N force acts on block 2 from the right, and the surface can be considered frictionless. Calculate:



(a) The acceleration of the system

Solution: The acceleration of the system is $-9\,\mathrm{ms}^{-2}$

(b) The acceleration of each block

Solution: Each block has an acceleration of $-9\,\mathrm{ms}^{-2}$

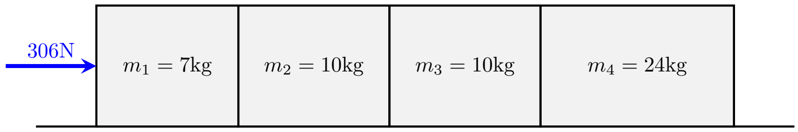
(c) The net force on each block

Solution: Block 1 has a net force of -90N
Block 2 has a net force of -36N

(d) The contact forces for each block

Solution: Block 1 has 1 force acting on it. They are:
-90N Contact Force from block 2
Block 2 has 2 forces acting on it. They are:
-126N Applied Force, 90N Contact Force from block 1

9. (6 points) The diagram shows 4 blocks in contact on a smooth surface. A 306N force acts on block 1 from the left, and the surface can be considered frictionless. Calculate:



(a) The acceleration of the system

> **Solution:** The acceleration of the system is $6\,\mathrm{ms}^{-2}$

(b) The acceleration of each block

> **Solution:** Each block has an acceleration of $6\,\mathrm{ms}^{-2}$

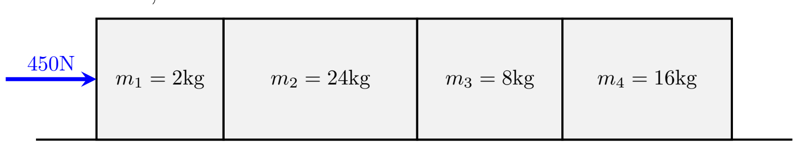
(c) The net force on each block

> **Solution:** Block 1 has a net force of 42N
> Block 2 has a net force of 60N
> Block 3 has a net force of 60N
> Block 4 has a net force of 144N

(d) The contact forces for each block

> **Solution:** Block 1 has 2 forces acting on it. They are:
> 306N Applied Force, -264N Contact Force from block 2
> Block 2 has 2 forces acting on it. They are:
> 264N Contact Force from block 1, -204N Contact Force from block 3
> Block 3 has 2 forces acting on it. They are:
> 204N Contact Force from block 2, -144N Contact Force from block 4
> Block 4 has 1 force acting on it. They are:
> 144N Contact Force from block 3

10. (6 points) The diagram shows 4 blocks in contact on a smooth surface. A 450N force acts on block 1 from the left, and the surface can be considered frictionless. Calculate:



(a) The acceleration of the system

> **Solution:** The acceleration of the system is $9\,\mathrm{ms}^{-2}$

(b) The acceleration of each block

> **Solution:** Each block has an acceleration of $9\,\mathrm{ms}^{-2}$

(c) The net force on each block

Solution: Block 1 has a net force of 18N
Block 2 has a net force of 216N
Block 3 has a net force of 72N
Block 4 has a net force of 144N

(d) The contact forces for each block

Solution: Block 1 has 2 forces acting on it. They are:
450N Applied Force, -432N Contact Force from block 2
Block 2 has 2 forces acting on it. They are:
432N Contact Force from block 1, -216N Contact Force from block 3
Block 3 has 2 forces acting on it. They are:
216N Contact Force from block 2, -144N Contact Force from block 4
Block 4 has 1 force acting on it. They are:
144N Contact Force from block 3